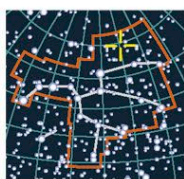


Sb SPIRAL GALAXY

Bode's Galaxy



CATALOG NUMBERS
M81, NGC 3031

DISTANCE
10.5 million light-years

DIAMETER
95,000 light-years

MAGNITUDE 6.9

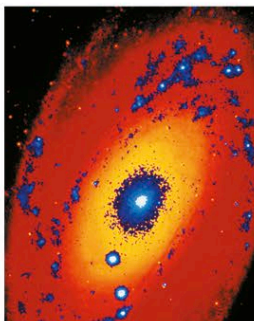
URSA MAJOR

Bode's Galaxy, also known as M81, is one of the brightest spiral galaxies visible from the Northern Hemisphere. It is the dominant member of a galaxy group lying near the Local Group. The galaxy is named after Johann Elert Bode, a German astronomer who found it in 1774.

Bode's Galaxy has had a close encounter with M82, the Cigar Galaxy (see below),

CLUSTERS REVEALED

This combined visible and ultraviolet image shows the hottest and brightest star clusters (blue and white blobs), lying in the core and spiral arms.



X-RAY SOURCES

A Chandra X-ray image shows a strong X-ray source at the galaxy's core surrounded by smaller sources, probably X-ray binary stars.

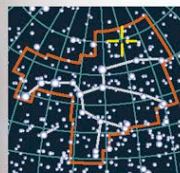
in the past few tens of millions of years. The near miss created tidal forces that enhanced the density waves (see p.303) in M81. The rate of star birth around the density waves increased, highlighting the spiral arms.

A long, straight dust lane along one side of the core could also have been created in the encounter.

By measuring the Doppler shifts of light from either side of the core, astronomers have found that the outer regions rotate more slowly than in most galaxies. This suggests that M81 has little of the dark matter that creates higher rotation rates in other galaxies.

IRREGULAR DISK GALAXY

Cigar Galaxy



CATALOG NUMBERS M82, NGC 3034

DISTANCE
12 million light-years

DIAMETER
40,000 light-years

MAGNITUDE 8.9

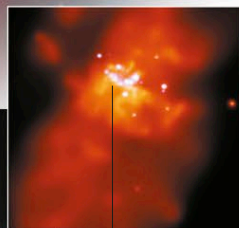
URSA MAJOR

The brightest and most spectacular example of a "starburst galaxy," the Cigar Galaxy (M82) is an irregularly shaped cloud of stars that looks like a cigar from Earth. It is undergoing a period of intense star birth as a result of a close encounter with Bode's Galaxy (M81). The near miss has disrupted the galaxy's center, creating the dark dust lanes that obscure much of the core and triggering the creation of many massive, brilliant star clusters in an area a few thousand light-years across.

At infrared wavelengths, M82 is the brightest galaxy in the sky. In 2014, astronomers observed one of the closest recent supernovae to Earth here.

STARBURST GALAXY

The intense activity in M82's core is luminous at optical and X-ray wavelengths. The young stars illuminate the nebulae with visible light, while those that have rapidly completed their life cycle form active black holes, emitting X-rays.



X-RAY VIEW

cluster of active black holes



OPTICAL IMAGE

Sb SPIRAL GALAXY

Black Eye Galaxy



CATALOG NUMBERS
M64, NGC 4826

DISTANCE
19 million light-years

DIAMETER
51,000 light-years

MAGNITUDE 8.5

COMA BERENICES

This distinctive galaxy has a dark dust lane, running in front of its core, from which it gets its name. The dust lane is unusual because it arcs above the galaxy's core in an orbit of its own. Because it has not yet settled into the plane of the galaxy's rotation, it must have a recent origin and probably dates from the galaxy's absorption of a smaller galaxy that strayed too close. Another bizarre feature of the Black Eye Galaxy is that its outer regions are rotating in the opposite direction of the inner regions. This could be another effect of the collision.



M64'S CENTRAL REGION AND DUST LANE